

Care Commons EHR

ONC Health IT Certification
Software Manual

2015 Edition Cures Update

Care Commons Consortium

October 21, 2025

Contents

I	Base EHR Certification	5
	Overview	6
1	170.315(a)(1) - CPOE Medications	7
1.1	Criterion Information	8
1.1.1	Maturity Level	8
1.1.2	Regulatory Text	8
1.2	Behavior-Driven Development Specification	8
1.3	Screenshots	9
2	170.315(a)(2) - CPOE Laboratory	10
2.1	Criterion Information	11
2.1.1	Maturity Level	11
2.1.2	Regulatory Text	11
2.2	Behavior-Driven Development Specification	11
2.3	Screenshots	12
3	170.315(a)(3) - CPOE Diagnostic Imaging	13
3.1	Criterion Information	14
3.1.1	Maturity Level	14
3.1.2	Regulatory Text	14
3.2	Behavior-Driven Development Specification	14
3.3	Screenshots	15
4	170.315(a)(4) - Drug Interaction Checks	16
4.1	Criterion Information	17
4.1.1	Maturity Level	17
4.1.2	Regulatory Text	17
4.2	Behavior-Driven Development Specification	17
4.3	Screenshots	19
5	170.315(a)(5) - Demographics	21
5.1	Criterion Information	22
5.1.1	Maturity Level	22
5.1.2	Regulatory Text	22
5.2	Behavior-Driven Development Specification	22
5.3	Screenshots	28
6	170.315(a)(9) - Clinical Decision Support (EXPIRED)	29

6.1	Criterion Information	30
6.1.1	Maturity Level	30
6.1.2	Expiration Notice	30
6.1.3	Regulatory Text	30
6.2	Behavior-Driven Development Specification	30
6.3	Algorithms	35
6.3.1	CDS Intervention Triggering	35
6.3.2	Intervention Configuration	35
6.3.3	Data-Driven Intervention Selection	35
6.3.4	Linked Referential CDS (Infobutton)	35
6.3.5	Source Attribution	35
6.4	Screenshots	36
7	170.315(b)(11) - Decision Support Interventions	37
7.1	Criterion Information	38
7.1.1	Maturity Level	38
7.1.2	Replacement Notice	38
7.1.3	Regulatory Text	38
7.2	Behavior-Driven Development Specification	38
7.3	Algorithms	43
7.3.1	Enhanced DSI Framework	43
7.3.2	Predictive DSI Algorithm Model	43
7.3.3	Fairness Metric	43
7.3.4	Demographic Transparency Function	43
7.3.5	Intervention Feedback Model	44
7.3.6	Validity and Fairness Monitoring	44
7.4	Screenshots	44
8	170.315(a)(14) - Implantable Device List	45
8.1	Criterion Information	46
8.1.1	Maturity Level	46
8.1.2	Regulatory Text	46
8.2	Behavior-Driven Development Specification	46
8.3	Algorithms	50
8.3.1	UDI Parsing Algorithm	50
8.3.2	GUDID Lookup Function	50
8.3.3	Device Status State Machine	51
8.3.4	Patient Device List	51
8.4	Screenshots	51
9	170.315(e)(1) - View, Download, Transmit	52
9.1	Criterion Information	53
9.1.1	Maturity Level	53
9.1.2	Regulatory Text	53
9.2	Behavior-Driven Development Specification	53
9.3	Algorithms	54
9.3.1	Patient Data Access Domain	54
9.3.2	View Function	54

9.3.3	Download Function	54
9.3.4	Transmission Function	54
9.3.5	Activity Log	55
9.3.6	Consent Management	55
9.4	Screenshots	55
10	170.315(g)(7) - Application Access - Patient Selection	56
10.1	Criterion Information	57
10.1.1	Maturity Level	57
10.1.2	Regulatory Text	57
10.2	Behavior-Driven Development Specification	57
10.3	Algorithms	58
10.3.1	Patient Identification Function	58
10.3.2	Token Generation	58
10.3.3	FHIR Patient Search	58
10.4	Screenshots	59
11	170.315(g)(9) - Application Access - All Data	60
11.1	Criterion Information	61
11.1.1	Maturity Level	61
11.1.2	Regulatory Text	61
11.2	Behavior-Driven Development Specification	61
11.3	Algorithms	62
11.3.1	All Data Request Function	62
11.3.2	USCDI Completeness	62
11.3.3	Date Filtering	62
11.4	Screenshots	63
12	170.315(g)(10) - Standardized API	64
12.1	Criterion Information	65
12.1.1	Maturity Level	65
12.1.2	Regulatory Text	65
12.2	Behavior-Driven Development Specification	65
12.3	Algorithms	66
12.3.1	FHIR API Endpoint Space	66
12.3.2	OAuth 2.0 Authorization	66
12.3.3	SMART Scopes	66
12.3.4	Token Introspection	67
12.3.5	Refresh Token Validity	67
12.3.6	Authorization Revocation	67
12.3.7	Bulk Data Export	67
12.4	Screenshots	67
A	Acronyms and Abbreviations	69
B	References	71
C	Certification Testing Notes	72
C.1	Inferno Testing	72

C.2 Test Execution	72
------------------------------	----

Part I

Base EHR Certification

Overview

This Part documents the Base EHR certification criteria per the ONC 2015 Edition Cures Update certification program. Base EHR requires **11 core criteria** to be met.

Certification Criteria Summary

Criterion	Title	Status
170.315(a)(1)	CPOE - Medications	Implemented
170.315(a)(2)	CPOE - Laboratory	Implemented
170.315(a)(3)	CPOE - Diagnostic Imaging	Implemented
170.315(a)(4)	Drug Interaction Checks	Implemented
170.315(a)(5)	Demographics	Implemented
170.315(a)(9)	Clinical Decision Support	Expired
170.315(b)(11)	Decision Support Interventions	Implemented
170.315(a)(14)	Implantable Device List	Implemented
170.315(e)(1)	View, Download, Transmit	Implemented
170.315(g)(7)	Application Access - Patient Selection	Implemented
170.315(g)(9)	Application Access - All Data	Implemented
170.315(g)(10)	Standardized API	Implemented

Table 1: Base EHR Certification Criteria

Note on 170.315(a)(9): This criterion expired January 1, 2025 and was replaced by 170.315(b)(11). Due to government website unavailability as of January 2025, it is unclear whether Base EHR requires the expired a-9 (grandfathered) or the replacement b-11. This manual documents both criteria.

Chapter 1

170.315(a)(1) - CPOE Medications

1.1 Criterion Information

1.1.1 Maturity Level

IMPLEMENTED

1.1.2 Regulatory Text

From 45 CFR 170.315(a)(1):

- (1) **Computerized provider order entry medications.**
 - (i) Enable a user to record, change, and access medication orders.
 - (ii) Optional. Include a “reason for order” field.

1.2 Behavior-Driven Development Specification

Listing 1.1: 170.315(a)(1) - CPOE Medications

```
# /Volumes/SonicMagic/Code/honeycomb-public-release/certification/bdd/170.315-a-1-cpoe-
# medications.feature
# 170.315(a)(1) - Computerized Provider Order Entry - Medications

# REGULATORY TEXT FROM 45 CFR 170.315(a)(1)
#
# (1) Computerized provider order entry medications.
#
# (i) Enable a user to record, change, and access medication orders.
#
# (ii) Optional. Include a "reason for order" field.

@170.315-a-1 @cpoe @medications @clinical
Feature: Computerized Provider Order Entry - Medications
  As a prescribing provider
  I want to electronically create and manage medication orders
  So that I can reduce prescribing errors and improve patient safety

  Background:
    Given I am authenticated as a prescribing provider
    And I have appropriate privileges to order medications
    And a patient record is open

  Scenario: Create a new medication order
    Given the patient has no existing medication order for "Lisinopril_10mg"
    When I create a medication order for "Lisinopril_10mg_oral_tablet_once_daily"
    Then the system shall enable recording of the medication order
    And the order shall be stored in the patient's electronic record
    And the order shall be available for review and transmission

  Scenario: Change an existing medication order
    Given the patient has an active medication order for "Lisinopril_10mg_once_daily"
    When I change the medication order to "Lisinopril_20mg_once_daily"
    Then the system shall enable changing of the medication order
```

And the modified order shall be stored in the patient's electronic record
And the modification history shall be maintained

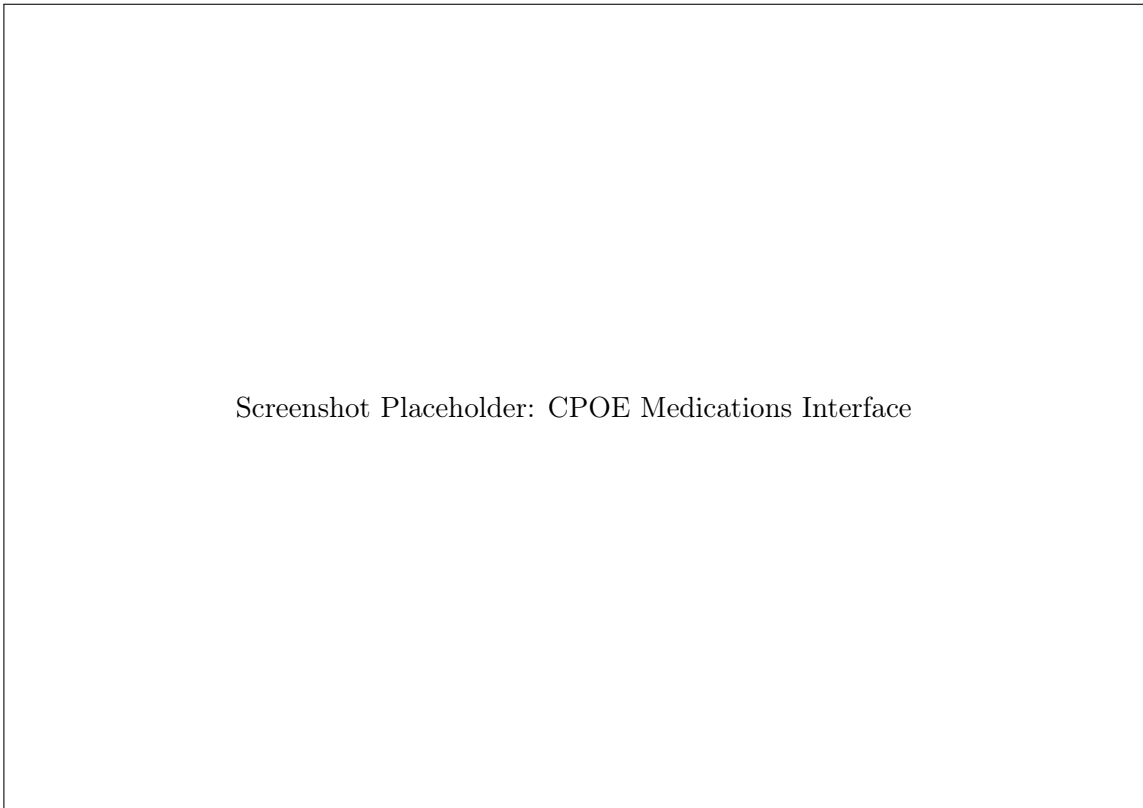
Scenario: Access existing medication orders

Given the patient has multiple active medication orders
When I request to view the patient's medication orders
Then the system shall enable access to all medication orders
And the orders shall be displayed with relevant details
And I shall be able to view order history

Scenario: Optionally include reason for medication order

Given I am creating a new medication order for "Metformin_500mg"
When I have the option to include a reason for the order
Then the system may include a "reason_for_order" field
And if provided, the reason shall be stored with the order

1.3 Screenshots



Screenshot Placeholder: CPOE Medications Interface

Figure 1.1: CPOE Medications - Order Entry Interface

Chapter 2

170.315(a)(2) - CPOE Laboratory

2.1 Criterion Information

2.1.1 Maturity Level

IMPLEMENTED

2.1.2 Regulatory Text

From 45 CFR 170.315(a)(2):

- (2) Computerized provider order entrylaboratory.
 - (i) Enable a user to record, change, and access laboratory orders.
 - (ii) Optional. Include a “reason for order” field.

2.2 Behavior-Driven Development Specification

Listing 2.1: 170.315(a)(2) - CPOE Laboratory

```
# /Volumes/SonicMagic/Code/honeycomb-public-release/certification/bdd/170.315-a-2-cpoe-
  laboratory.feature
# 170.315(a)(2) - Computerized Provider Order Entry - Laboratory

# REGULATORY TEXT FROM 45 CFR 170.315(a)(2)
#
# (2) Computerized provider order entrylaboratory.
#
# (i) Enable a user to record, change, and access laboratory orders.
#
# (ii) Optional. Include a "reason for order" field.

@170.315-a-2 @cpoe @laboratory @clinical
Feature: Computerized Provider Order Entry - Laboratory
  As an ordering provider
  I want to electronically create and manage laboratory orders
  So that I can efficiently order appropriate diagnostic tests

Background:
  Given I am authenticated as an ordering provider
  And I have appropriate privileges to order laboratory tests
  And a patient record is open

Scenario: Create a new laboratory order
  Given the patient requires a "Complete_Blood_Count"
  When I create a laboratory order for "CBC_with_differential"
  Then the system shall enable recording of the laboratory order
  And the order shall include all necessary details for specimen collection
  And the order shall be transmitted to the laboratory system

Scenario: Change an existing laboratory order
  Given the patient has a pending laboratory order for "Basic_Metabolic_Panel"
  When I change the laboratory order to "Comprehensive_Metabolic_Panel"
  Then the system shall enable changing of the laboratory order
```

And the modified order shall be communicated to the laboratory
And the change shall be documented in the order history

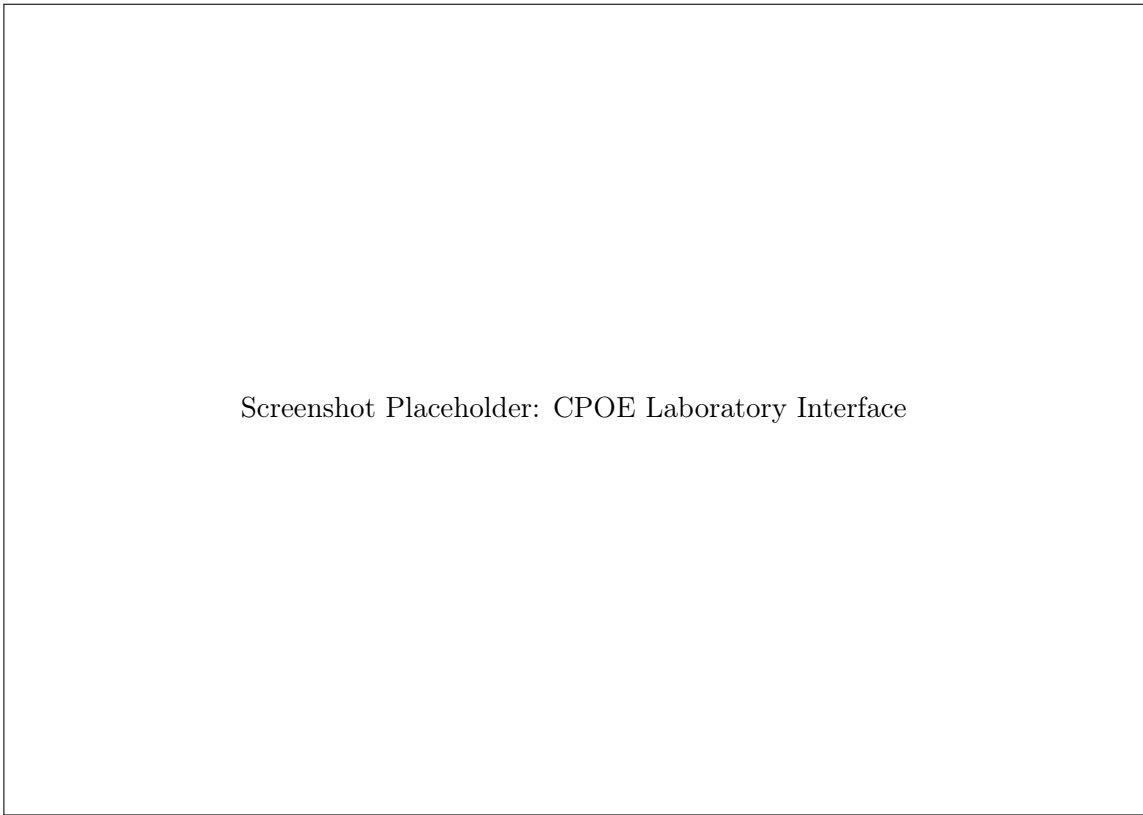
Scenario: Access existing laboratory orders

Given the patient has multiple laboratory orders in various states
When I request to view the patient's laboratory orders
Then the system shall enable access to all laboratory orders
And the orders shall show current status (pending, completed, cancelled)
And I shall be able to view results for completed orders

Scenario: Optionally include reason for laboratory order

Given I am creating a new laboratory order for "Hemoglobin_A1c"
When I have the option to include a reason for the order
Then the system may include a "reason_for_order" field
And if provided, the reason shall be stored with the order

2.3 Screenshots



Screenshot Placeholder: CPOE Laboratory Interface

Figure 2.1: CPOE Laboratory - Order Entry Interface

Chapter 3

170.315(a)(3) - CPOE Diagnostic Imaging

3.1 Criterion Information

3.1.1 Maturity Level

IMPLEMENTED

3.1.2 Regulatory Text

From 45 CFR 170.315(a)(3):

- (3) Computerized provider order entry diagnostic imaging.
 - (i) Enable a user to record, change, and access diagnostic imaging orders.
 - (ii) Optional. Include a “reason for order” field.

3.2 Behavior-Driven Development Specification

Listing 3.1: 170.315(a)(3) - CPOE Diagnostic Imaging

```
# /Volumes/SonicMagic/Code/honeycomb-public-release/certification/bdd/170.315-a-3-cpoe-
  diagnostic-imaging.feature
# 170.315(a)(3) - Computerized Provider Order Entry - Diagnostic Imaging

# REGULATORY TEXT FROM 45 CFR 170.315(a)(3)
#
# (3) Computerized provider order entry diagnostic imaging.
#
# (i) Enable a user to record, change, and access diagnostic imaging orders.
#
# (ii) Optional. Include a "reason for order" field.

@170.315-a-3 @cpoe @diagnostic-imaging @clinical
Feature: Computerized Provider Order Entry - Diagnostic Imaging
  As an ordering provider
  I want to electronically create and manage diagnostic imaging orders
  So that I can order appropriate imaging studies efficiently

Background:
  Given I am authenticated as an ordering provider
  And I have appropriate privileges to order diagnostic imaging
  And a patient record is open

Scenario: Create a new diagnostic imaging order
  Given the patient requires a chest X-ray
  When I create a diagnostic imaging order for "Chest_X-ray_PA_and_lateral"
  Then the system shall enable recording of the diagnostic imaging order
  And the order shall include imaging modality and anatomical location
  And the order shall be transmitted to the radiology system

Scenario: Change an existing diagnostic imaging order
  Given the patient has a pending imaging order for "CT_abdomen_without_contrast"
  When I change the imaging order to "CT_abdomen_and_pelvis_with_contrast"
  Then the system shall enable changing of the diagnostic imaging order
```

And the modified order shall be communicated to radiology
And the modification shall be documented

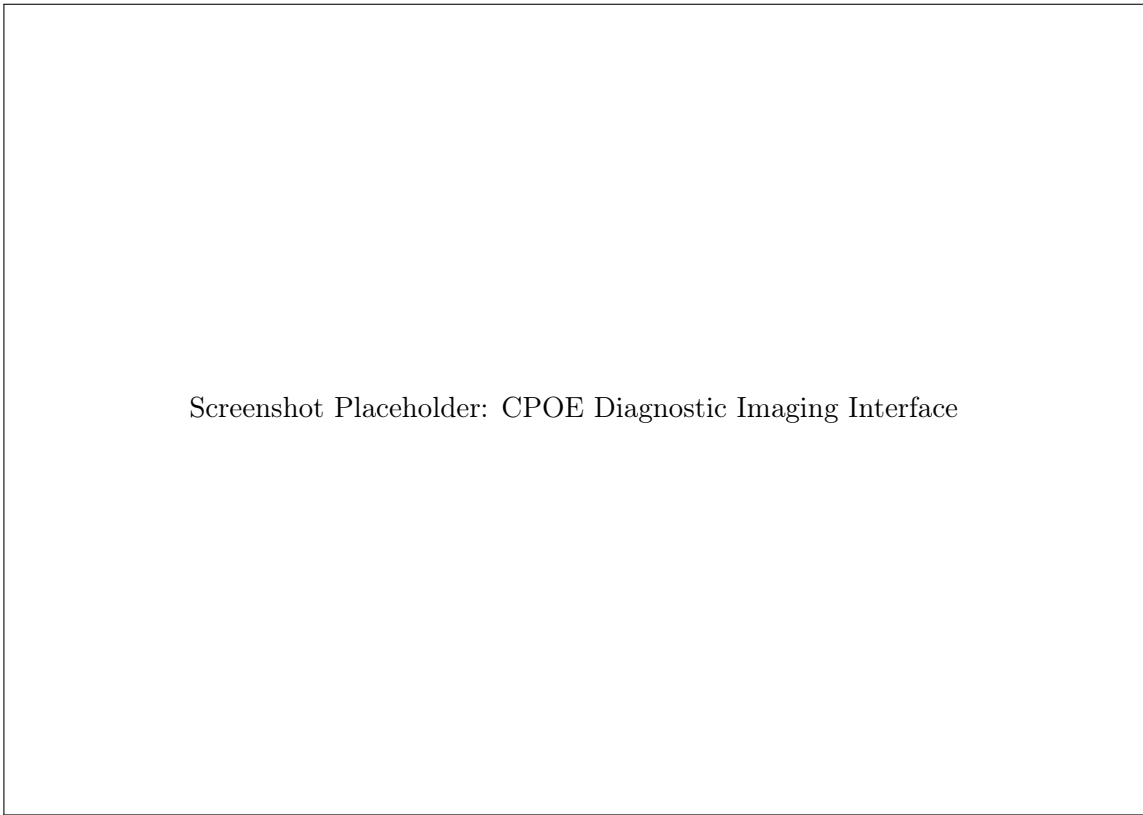
Scenario: Access existing diagnostic imaging orders

Given the patient has multiple imaging orders and completed studies
When I request to view the patient's diagnostic imaging orders
Then the system shall enable access to all diagnostic imaging orders
And the orders shall show current status
And I shall be able to access imaging reports and images

Scenario: Optionally include reason for diagnostic imaging order

Given I am creating a new imaging order for "MRI_lumbar_spine"
When I have the option to include a reason for the order
Then the system may include a "reason_for_order" field
And if provided, the reason shall support appropriate utilization

3.3 Screenshots



Screenshot Placeholder: CPOE Diagnostic Imaging Interface

Figure 3.1: CPOE Diagnostic Imaging - Order Entry Interface

Chapter 4

170.315(a)(4) - Drug Interaction Checks

4.1 Criterion Information

4.1.1 Maturity Level

IMPLEMENTED

4.1.2 Regulatory Text

From 45 CFR 170.315(a)(4):

(4) Drug-drug, drug-allergy interaction checks for CPOE

(i) Interventions. Before a medication order is completed and acted upon during computerized provider order entry (CPOE), interventions must automatically indicate to a user drug-drug and drug-allergy contraindications based on a patient's medication list and medication allergy list.

(ii) Adjustments.

(A) Enable the severity level of interventions provided for drug-drug interaction checks to be adjusted.

(B) Limit the ability to adjust severity levels in at least one of these two ways:

(1) To a specific set of identified users.

(2) As a system administrative function.

4.2 Behavior-Driven Development Specification

Listing 4.1: 170.315(a)(4) - Drug Interaction Checks

```
# /Volumes/SonicMagic/Code/honeycomb-public-release/certification/bdd/170.315-a-4-drug-
interactions.feature
# 170.315(a)(4) - Drug-Drug, Drug-Allergy Interaction Checks for CPOE

# REGULATORY TEXT FROM 45 CFR 170.315(a)(4)
#
# (4) Drug-drug, drug-allergy interaction checks for CPOE
#
# (i) Interventions. Before a medication order is completed and acted upon during
computerized provider order entry (CPOE), interventions must automatically indicate
to a user drug-drug and drug-allergy contraindications based on a patient's
medication list and medication allergy list.
#
# (ii) Adjustments.
#
# (A) Enable the severity level of interventions provided for drug-drug interaction
checks to be adjusted.
#
# (B) Limit the ability to adjust severity levels in at least one of these two ways:
#
# (1) To a specific set of identified users.
#
# (2) As a system administrative function.

@170.315-a-4 @cpoe @drug-interactions @safety @clinical
```

Feature: Drug-Drug and Drug-Allergy Interaction Checks for CPOE

As a prescribing provider

I want automatic alerts for potential drug interactions

So that I can prevent adverse drug events and improve patient safety

Background:

Given I am authenticated as a prescribing provider

And the patient has an active medication list

And the patient has a medication allergy list

And interaction checking is enabled

Scenario: Automatic drug-drug interaction checking during CPOE

Given the patient is currently taking "Warfarin_5mg_daily"

When I attempt to order "Aspirin_325mg_daily"

And I complete the medication order entry

But before the order is completed and acted upon

Then the system shall automatically indicate drug-drug contraindications

And the intervention shall display before order completion

And the alert shall describe the interaction between Warfarin and Aspirin

Scenario: Automatic drug-allergy interaction checking during CPOE

Given the patient has a documented allergy to "Penicillin"

When I attempt to order "Amoxicillin_500mg"

And I complete the medication order entry

But before the order is completed and acted upon

Then the system shall automatically indicate drug-allergy contraindications

And the intervention shall display before order completion

And the alert shall identify the cross-reactivity concern

Scenario: Display interventions based on active medication list

Given the patient's active medication list contains multiple medications

When I order a new medication

Then the system shall check against all medications in the active medication list

And all relevant drug-drug interactions shall be displayed

Scenario: Display interventions based on medication allergy list

Given the patient's allergy list contains documented allergies

When I order a new medication

Then the system shall check against all allergies in the medication allergy list

And all relevant drug-allergy contraindications shall be displayed

Scenario: Enable severity level adjustment for drug-drug interactions

Given I am a system administrator or authorized user

When I access the drug-drug interaction settings

Then the system shall enable adjustment of severity levels

And severity levels shall control which interactions trigger alerts

Scenario: Limit severity level adjustment to authorized users

Given severity level adjustment capability exists

When limiting who can adjust severity levels

Then the system shall limit adjustment to a specific set of identified users

Or the system shall limit adjustment as a system administrative function

And unauthorized users shall not be able to modify severity settings

Scenario: Display high-severity drug-drug interaction

Given the interaction checking system is configured with severity levels

And the patient is taking "MAO_Inhibitor"

When I attempt to order "SSRI_antidepressant"

Then the system shall display high-severity interaction alert

And the alert shall be prominent and require acknowledgment

Scenario: Display moderate-severity drug-drug interaction

Given the interaction checking system is configured with severity levels

And the patient is taking "Atorvastatin"

When I attempt to order "Clarithromycin"

Then the system shall display moderate-severity interaction alert

And the alert shall provide clinical guidance

Scenario: Provider override of drug interaction alert

Given a drug-drug interaction alert is displayed

When I review the alert and determine the benefit outweighs risk

Then the system shall allow me to override the alert

And the override shall require documentation of clinical reasoning

And the override shall be recorded in the audit log

Scenario: Multiple simultaneous interaction alerts

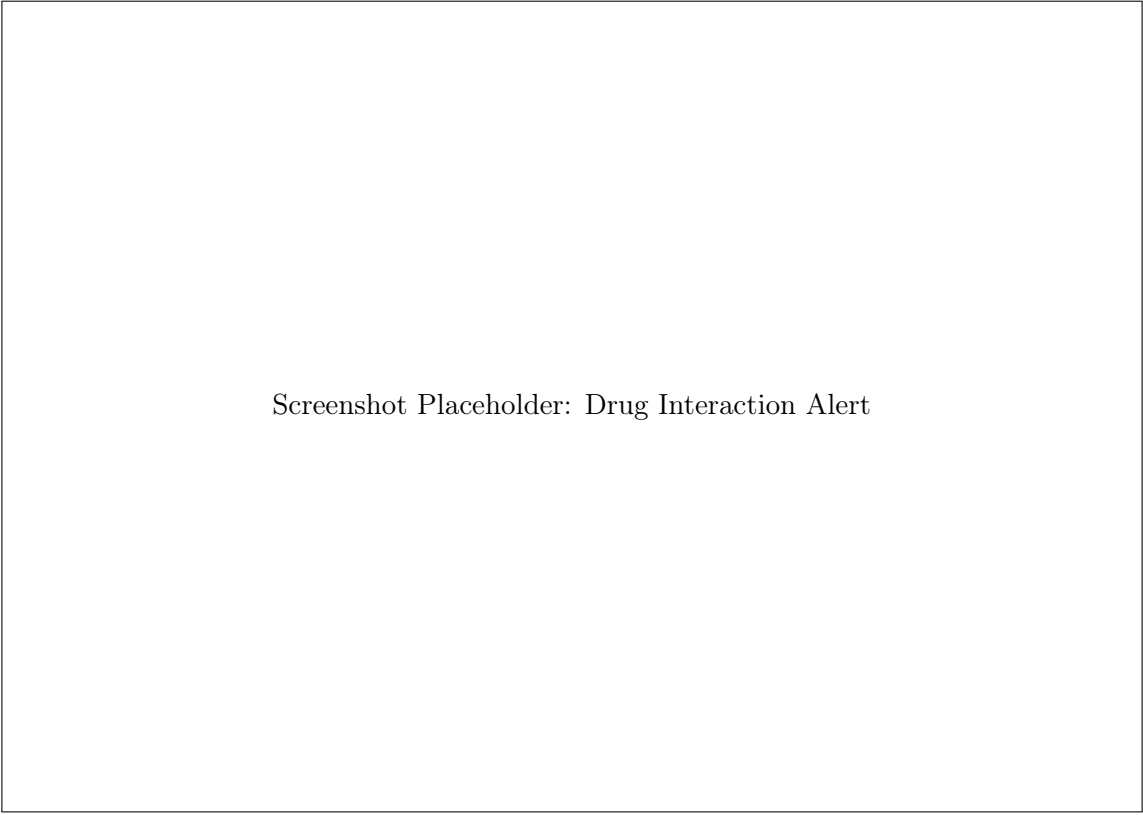
Given the patient has complex medication regimen

When I order a medication that interacts with multiple existing medications

Then the system shall display all relevant interactions

And interactions shall be prioritized by severity

4.3 Screenshots



Screenshot Placeholder: Drug Interaction Alert

Figure 4.1: Drug-Drug Interaction Alert Display

Chapter 5

170.315(a)(5) - Demographics

5.1 Criterion Information

5.1.1 Maturity Level

IMPLEMENTED

5.1.2 Regulatory Text

From 45 CFR 170.315(a)(5):

(5) Patient demographics and observations.

(i) Enable a user to record, change, and access patient demographic and observations data including race, ethnicity, preferred language, sex, sex parameter for clinical use, sexual orientation, gender identity, name to use, pronouns, and date of birth.

(Full regulatory text abbreviated for brevity. See BDD specification for complete requirements.)

5.2 Behavior-Driven Development Specification

Listing 5.1: 170.315(a)(5) - Demographics

```
# /Volumes/SonicMagic/Code/honeycomb-public-release/certification/bdd/170.315-a-5-
# demographics-observations.feature
# 170.315(a)(5) - Patient Demographics and Observations

# REGULATORY TEXT FROM 45 CFR 170.315(a)(5)
#
# (5) Patient demographics and observations.
#
# (i) Enable a user to record, change, and access patient demographic and observations
# data including race, ethnicity, preferred language, sex, sex parameter for clinical
# use, sexual orientation, gender identity, name to use, pronouns, and date of birth.
#
# (A) Race and ethnicity.
#
# (1) Enable each one of a patient's races to be recorded in accordance with, at a
# minimum, the standard specified in 170.207(f)(3) and whether a patient declines to
# specify race.
#
# (2) Enable each one of a patient's ethnicities to be recorded in accordance with, at a
# minimum, the standard specified in 170.207(f)(3) and whether a patient declines to
# specify ethnicity.
#
# (3) Aggregate each one of the patient's races and ethnicities recorded in accordance
# with paragraphs (a)(5)(i)(A)(1) and (2) of this section to the categories in the
# standard specified in 170.207(f)(1).
#
# (B) Preferred language. Enable preferred language to be recorded in accordance with the
# standard specified in 170.207(g)(2) and whether a patient declines to specify a
# preferred language.
#
```

```

# (C) Sex. Enable sex to be recorded in accordance with the standard specified in
    170.207(n)(1) for the period up to and including December 31, 2025; or 170.207(n)(2)
.
#
# (D) Sexual orientation. Enable sexual orientation to be recorded in accordance with, at
    a minimum, the version of the standard specified in 170.207(o)(1) for the period up
    to and including December 31, 2025; or 170.207(o)(3), as well as whether a patient
    declines to specify sexual orientation.
#
# (E) Gender identity. Enable gender identity to be recorded in accordance with, at a
    minimum, the version of the standard specified in 170.207(o)(2) for the period up to
    and including December 31, 2025; or 170.207(o)(3), as well as whether a patient
    declines to specify gender identity.
#
# (F) Sex Parameter for Clinical Use. Enable at least one sex parameter for clinical use
    to be recorded in accordance with, at a minimum, the version of the standard
    specified in 170.207(n)(3). Conformance with this paragraph is required by January
    1, 2026.
#
# (G) Name to Use. Enable at least one preferred name to use to be recorded. Conformance
    with this paragraph is required by January 1, 2026.
#
# (H) Pronouns. Enable at least one pronoun to be recorded in accordance with, at a
    minimum, the version of the standard specified in 170.207(o)(4). Conformance with
    this paragraph is required by January 1, 2026.
#
# (ii) Inpatient setting only. Enable a user to record, change, and access the
    preliminary cause of death and date of death in the event of mortality.

@170.315-a-5 @demographics @patient-data @clinical
Feature: Patient Demographics and Observations
As a healthcare provider
I want to record comprehensive patient demographic and observation data
So that I can provide culturally competent and personalized care

Background:
Given I am authenticated as a healthcare provider
And I have appropriate privileges to record patient demographics
And a patient record is open or being created

# ----- Race and Ethnicity -----

Scenario: Record patient's race using standard vocabulary
Given I am recording patient demographics
When I record the patient's race
Then the system shall enable recording per 170.207(f)(3) standard
And the system shall support granular race categories
And the system shall enable recording multiple races

Scenario: Record multiple races for a patient
Given a patient identifies with multiple racial categories
When I record the patient's races
Then the system shall enable each one of the patient's races to be recorded
And each race shall be stored as discrete data element

```

And all recorded races shall be retrievable

Scenario: Record patient decline to specify race

Given I am recording patient demographics

When the patient declines to specify their race

Then the system shall enable recording that patient declines to specify race

And this shall be stored as distinct from missing data

Scenario: Record patient's ethnicity using standard vocabulary

Given I am recording patient demographics

When I record the patient's ethnicity

Then the system shall enable recording per 170.207(f)(3) standard

And the system shall support Hispanic/Latino ethnicity categories

Scenario: Record multiple ethnicities for a patient

Given a patient identifies with multiple ethnic categories

When I record the patient's ethnicities

Then the system shall enable each one of the patient's ethnicities to be recorded

And each ethnicity shall be stored as discrete data element

Scenario: Record patient decline to specify ethnicity

Given I am recording patient demographics

When the patient declines to specify their ethnicity

Then the system shall enable recording that patient declines to specify ethnicity

And this shall be stored as distinct from missing data

Scenario: Aggregate race and ethnicity to standard categories

Given granular race and ethnicity data has been recorded

When aggregating for reporting purposes

Then the system shall aggregate to categories per 170.207(f)(1)

And aggregation shall maintain data integrity

And original granular data shall be preserved

----- Preferred Language -----

Scenario: Record patient's preferred language

Given I am recording patient demographics

When I record the patient's preferred language

Then the system shall enable recording per 170.207(g)(2) standard

And the system shall use ISO 639-2 language codes

Scenario: Record patient decline to specify preferred language

Given I am recording patient demographics

When the patient declines to specify preferred language

Then the system shall enable recording that patient declines to specify

And this shall be stored as distinct from missing data

----- Sex -----

Scenario: Record patient's sex (until December 31, 2025)

Given the current date is before December 31, 2025

When I record the patient's sex

Then the system shall enable recording per 170.207(n)(1) standard

And the system shall support administrative sex categories

Scenario: Record patient's sex (after December 31, 2025)

Given the current date is after December 31, 2025

When I record the patient's sex

Then the system shall enable recording per 170.207(n)(2) standard

And the system shall support updated sex categories

----- Sex Parameter for Clinical Use (Required by January 1, 2026) -----

Scenario: Record sex parameter for clinical use

Given the current date is on or after January 1, 2026

When I record clinical observations requiring sex parameter

Then the system shall enable recording at least one sex parameter for clinical use

And the recording shall be per 170.207(n)(3) standard

And the parameter shall be context-specific for clinical use

Scenario: Support multiple sex parameters for clinical use

Given the current date is on or after January 1, 2026

And different clinical contexts require different parameters

When I record sex parameters for various clinical purposes

Then the system shall enable recording multiple sex parameters

And each parameter shall be associated with its clinical context

----- Sexual Orientation -----

Scenario: Record patient's sexual orientation (until December 31, 2025)

Given the current date is before December 31, 2025

When I record the patient's sexual orientation

Then the system shall enable recording per 170.207(o)(1) standard

And the system shall use LOINC and SNOMED CT codes

Scenario: Record patient's sexual orientation (after December 31, 2025)

Given the current date is after December 31, 2025

When I record the patient's sexual orientation

Then the system shall enable recording per 170.207(o)(3) standard

And the system shall support updated vocabulary

Scenario: Record patient decline to specify sexual orientation

Given I am recording patient demographics

When the patient declines to specify sexual orientation

Then the system shall enable recording that patient declines to specify

And this shall be stored as distinct from missing data

----- Gender Identity -----

Scenario: Record patient's gender identity (until December 31, 2025)

Given the current date is before December 31, 2025

When I record the patient's gender identity

Then the system shall enable recording per 170.207(o)(2) standard

And the system shall use standard gender identity vocabulary

Scenario: Record patient's gender identity (after December 31, 2025)

Given the current date is after December 31, 2025

When I record the patient's gender identity

Then the system shall enable recording per 170.207(o)(3) standard
And the system shall support updated vocabulary

Scenario: Record patient decline to specify gender identity

Given I am recording patient demographics
When the patient declines to specify gender identity
Then the system shall enable recording that patient declines to specify
And this shall be stored as distinct from missing data

----- *Name to Use (Required by January 1, 2026)* -----

Scenario: Record patient's preferred name to use

Given the current date is on or after January 1, 2026
When I record the patient's preferred name
Then the system shall enable recording at least one preferred name to use
And the preferred name shall be distinct from legal name
And the preferred name shall be usable in clinical workflows

Scenario: Display patient's preferred name in user interface

Given the current date is on or after January 1, 2026
And the patient has a preferred name recorded
When displaying patient information
Then the system should prominently display the preferred name
And the legal name shall remain accessible when needed

----- *Pronouns (Required by January 1, 2026)* -----

Scenario: Record patient's pronouns

Given the current date is on or after January 1, 2026
When I record the patient's pronouns
Then the system shall enable recording at least one pronoun
And the recording shall be per 170.207(o)(4) standard
And the pronouns shall use standard vocabulary

Scenario: Support multiple pronoun sets

Given the current date is on or after January 1, 2026
And a patient uses different pronouns in different contexts
When I record the patient's pronouns
Then the system shall enable recording multiple pronoun sets
And each pronoun set shall be stored and retrievable

----- *Date of Birth* -----

Scenario: Record patient's complete date of birth

Given I am recording patient demographics
When I record the patient's date of birth
Then the system shall enable recording year, month, and day
And the date shall be stored in standard format
And the date shall be used for age calculations

Scenario: Handle unknown date of birth

Given the patient's exact date of birth is unknown
When I record demographic information
Then the system shall enable recording null value for unknown date of birth

And the system shall distinguish unknown from missing data

----- *Change and Access Demographics* -----

Scenario: Change patient demographic data

Given patient demographic data has been recorded

When demographic information changes or needs correction

Then the system shall enable changing all demographic data elements

And changes shall be audited with timestamp and user

And history of changes shall be maintained

Scenario: Access patient demographic data

Given patient demographic data has been recorded

When I need to view patient demographics

Then the system shall enable access to all demographic data elements

And demographic data shall be displayed clearly

And demographic data shall be available throughout clinical workflows

----- *Inpatient-Only: Mortality Data* -----

@inpatient-only

Scenario: Record preliminary cause of death in inpatient setting

Given I am authenticated in an inpatient setting

And a patient mortality has occurred

When I record mortality information

Then the system shall enable recording preliminary cause of death

And the cause shall be coded appropriately

And the information shall be available for death certificate

@inpatient-only

Scenario: Record date of death in inpatient setting

Given I am authenticated in an inpatient setting

And a patient mortality has occurred

When I record mortality information

Then the system shall enable recording date of death

And the date shall include date and time

And the date shall be stored in standard format

@inpatient-only

Scenario: Change mortality data in inpatient setting

Given mortality data has been recorded

When corrections are needed to mortality information

Then the system shall enable changing preliminary cause of death

And the system shall enable changing date of death

And all changes shall be audited

@inpatient-only

Scenario: Access mortality data in inpatient setting

Given mortality data has been recorded

When generating death certificate or reports

Then the system shall enable access to preliminary cause of death

And the system shall enable access to date of death

And the data shall be formatted appropriately for reporting

5.3 Screenshots

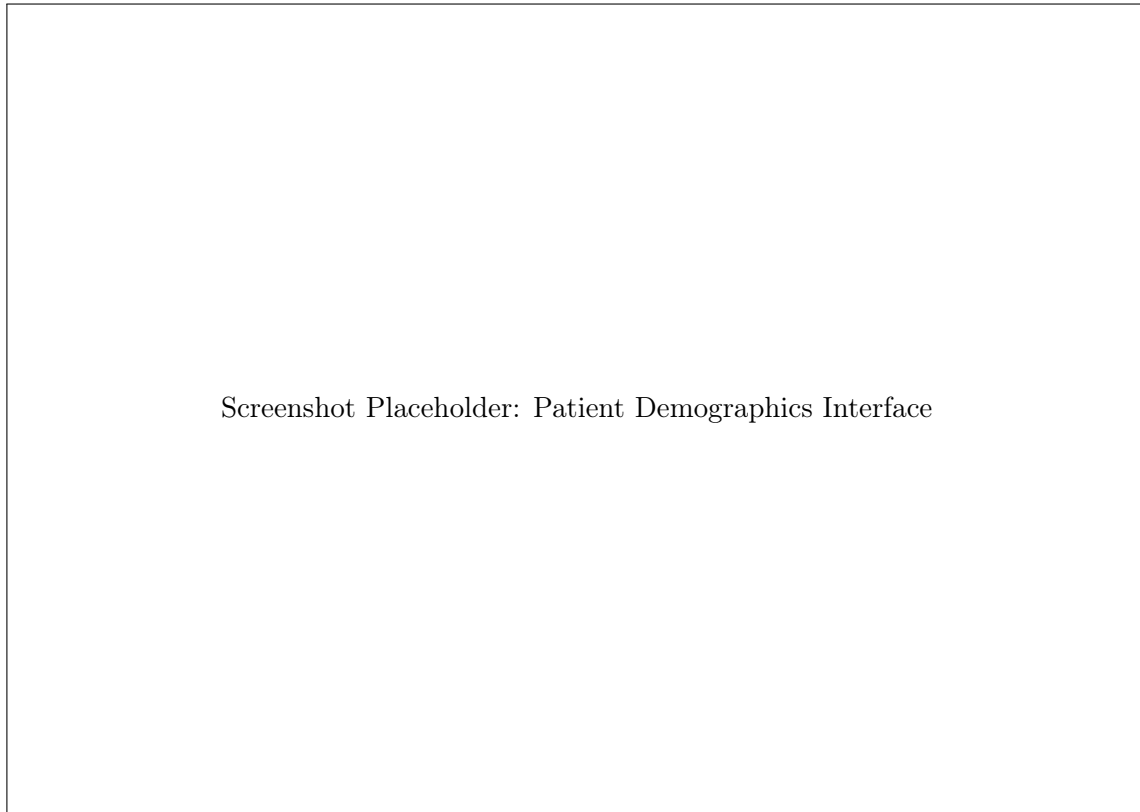


Figure 5.1: Patient Demographics - Data Entry Interface

Chapter 6

170.315(a)(9) - Clinical Decision Support (EXPIRED)

6.1 Criterion Information

6.1.1 Maturity Level

IMPLEMENTED (EXPIRED)

6.1.2 Expiration Notice

CRITICAL: This criterion EXPIRED on January 1, 2025 per 45 CFR 170.315(a)(9)(vi).

This criterion has been replaced by 170.315(b)(11) - Decision Support Interventions (see Chapter 7).

Due to government website unavailability as of January 2025, it is unclear whether Base EHR certifications issued before the expiration date can still use this criterion (grandfathered), or if all certifications must now use 170.315(b)(11).

6.1.3 Regulatory Text

From 45 CFR 170.315(a)(9):

(9) Clinical decision support (CDS)

(i) CDS intervention interaction. Interventions provided to a user must occur when a user is interacting with technology.

(Full regulatory text abbreviated. See BDD specification for complete requirements.)

(vi) Expiration of criterion. The adoption of this criterion for purposes of the ONC Health IT Certification Program expires on January 1, 2025.

6.2 Behavior-Driven Development Specification

Listing 6.1: 170.315(a)(9) - Clinical Decision Support

```
# /Volumes/SonicMagic/Code/honeycomb-public-release/certification/bdd/170.315-a-9-
  clinical-decision-support.feature
# 170.315(a)(9) - Clinical Decision Support (CDS)
# NOTE: This criterion expires January 1, 2025

# REGULATORY TEXT FROM 45 CFR 170.315(a)(9)
#
# (9) Clinical decision support (CDS)
#
# (i) CDS intervention interaction. Interventions provided to a user must occur when a
  user is interacting with technology.
#
# (ii) CDS configuration.
#
# (A) Enable interventions and reference resources specified in paragraphs (a)(9)(iii)
  and (iv) of this section to be configured by a limited set of identified users (e.g.,
  system administrator) based on a user's role.
#
# (B) Enable interventions:
#
# (1) Based on the following data:
```

```

#
# (i) Problem list;
#
# (ii) Medication list;
#
# (iii) Allergy and intolerance list;
#
# (iv) At least one demographic specified in paragraph (a)(5)(i) of this section;
#
# (v) Laboratory tests; and
#
# (vi) Vital signs.
#
# (2) When a patient's medications, allergies and intolerance, and problems are
#       incorporated from a transition of care/referral summary received and pursuant to
#       paragraph (b)(2)(iii)(D) of this section.
#
# (iii) Evidence-based decision support interventions. Enable a limited set of identified
#       users to select (i.e., activate) electronic CDS interventions (in addition to drug-
#       drug and drug-allergy contraindication checking) based on each one and at least one
#       combination of the data referenced in paragraphs (a)(9)(ii)(B)(1)(i) through (vi) of
#       this section.
#
# (iv) Linked referential CDS.
#
# (A) Identify for a user diagnostic and therapeutic reference information in accordance
#       at least one of the following standards and implementation specifications:
#
# (1) The standard and implementation specifications specified in 170.204(b)(3).
#
# (2) The standard and implementation specifications specified in 170.204(b)(4).
#
# (B) For paragraph (a)(9)(iv)(A) of this section, technology must be able to identify
#       for a user diagnostic or therapeutic reference information based on each one and at
#       least one combination of the data referenced in paragraphs (a)(9)(ii)(B)(1)(i), (ii),
#       and (iv) of this section.
#
# (v) Source attributes. Enable a user to review the attributes as indicated for all CDS
#       resources:
#
# (A) For evidence-based decision support interventions under paragraph (a)(9)(iii) of
#       this section:
#
# (1) Bibliographic citation of the intervention (clinical research/guideline);
#
# (2) Developer of the intervention (translation from clinical research/guideline);
#
# (3) Funding source of the intervention development technical implementation; and
#
# (4) Release and, if applicable, revision date(s) of the intervention or reference
#       source.
#
# (B) For linked referential CDS in paragraph (a)(9)(iv) of this section and drug-drug,
#       drug-allergy interaction checks in paragraph (a)(4) of this section, the developer of

```

the intervention, and where clinically indicated, the bibliographic citation of the intervention (clinical research/guideline).

#

(vi) Expiration of criterion. The adoption of this criterion for purposes of the ONC Health IT Certification Program expires on January 1, 2025.

@170.315-a-9 @cds @clinical-decision-support @expires-2025 @clinical

Feature: Clinical Decision Support (CDS)

As a healthcare provider

I want evidence-based clinical decision support

So that I can make informed treatment decisions

Background:

Given I am authenticated as a healthcare provider

And CDS is properly configured

And the current date is before January 1, 2025

----- CDS Intervention Interaction -----

Scenario: CDS interventions occur during user interaction

Given I am actively interacting with the health IT system

When clinical decision support is triggered

Then interventions shall be provided during my interaction with technology

And interventions shall not be delayed until after interaction ends

----- CDS Configuration -----

Scenario: Configure CDS interventions based on user role

Given I am a system administrator or authorized configurator

When I configure clinical decision support interventions

Then the system shall enable configuration by limited set of identified users

And configuration shall be based on user role

And configuration capabilities shall be restricted appropriately

Scenario: Configure CDS reference resources based on user role

Given I am a system administrator or authorized configurator

When I configure reference resources for CDS

Then the system shall enable configuration by limited set of identified users

And configuration shall be based on user role

----- CDS Based on Patient Data -----

Scenario: Enable interventions based on problem list

Given the patient has documented problems

When CDS evaluates patient data

Then the system shall enable interventions based on problem list data

Scenario: Enable interventions based on medication list

Given the patient has active medications

When CDS evaluates patient data

Then the system shall enable interventions based on medication list data

Scenario: Enable interventions based on allergy and intolerance list

Given the patient has documented allergies and intolerances

When CDS evaluates patient data
Then the system shall enable interventions based on allergy and intolerance list

Scenario: Enable interventions based on demographics

Given the patient has demographic data recorded
When CDS evaluates patient data
Then the system shall enable interventions based on at least one demographic
And demographic data may include age, sex, race, ethnicity, etc.

Scenario: Enable interventions based on laboratory tests

Given the patient has laboratory test results
When CDS evaluates patient data
Then the system shall enable interventions based on laboratory test data

Scenario: Enable interventions based on vital signs

Given the patient has vital signs recorded
When CDS evaluates patient data
Then the system shall enable interventions based on vital signs data

Scenario: Enable interventions from incorporated transition of care data

Given medications, allergies, and problems are incorporated from transition of care summary
When the data is incorporated per 170.315(b)(2)(iii)(D)
Then the system shall enable interventions based on incorporated data
And interventions shall trigger automatically upon incorporation

----- Evidence-Based Decision Support Interventions -----

Scenario: Select evidence-based CDS interventions

Given I am an authorized user selecting CDS interventions
When I activate electronic CDS interventions
Then the system shall enable selection based on each individual data element
And the system shall enable selection based on combinations of data elements
And data elements include problem list, medication list, allergies, demographics, lab tests, and vital signs

Scenario: Activate CDS intervention based on problem list

Given I am selecting CDS interventions
When I activate an intervention based on problem list
Then the intervention shall utilize problem list data
And the intervention shall provide evidence-based guidance

Scenario: Activate CDS intervention based on multiple data elements

Given I am selecting CDS interventions
When I activate an intervention based on combination of data elements
Then the intervention shall utilize all specified data elements
And the intervention shall provide integrated clinical guidance

----- Linked Referential CDS -----

Scenario: Identify diagnostic reference information via Infobutton

Given I need diagnostic reference information
When I access linked referential CDS
Then the system shall identify information per 170.204(b)(3) standard

And information shall be context-specific to clinical scenario

Scenario: Identify therapeutic reference information via Infobutton

Given I need therapeutic reference information

When I access linked referential CDS

Then the system shall identify information per 170.204(b)(4) standard

And information shall be context-specific to clinical scenario

Scenario: Identify reference information based on problem list

Given the patient has documented problems

When I access linked referential CDS

Then the system shall identify information based on problem list data

Scenario: Identify reference information based on medication list

Given the patient has active medications

When I access linked referential CDS

Then the system shall identify information based on medication list data

Scenario: Identify reference information based on demographics

Given the patient has demographic data

When I access linked referential CDS

Then the system shall identify information based on demographic data

Scenario: Identify reference information based on combination of data

Given multiple types of patient data are available

When I access linked referential CDS

Then the system shall identify information based on combinations of data

And combinations may include problem list, medications, and demographics

----- Source Attributes for CDS -----

Scenario: Display source attributes for evidence-based interventions

Given an evidence-based CDS intervention is active

When I review the intervention source information

Then the system shall enable review of bibliographic citation

And the system shall display developer of the intervention

And the system shall display funding source of intervention development

And the system shall display release and revision dates

Scenario: Display source attributes for drug interaction checks

Given drug-drug or drug-allergy interaction check is displayed

When I review the intervention source information

Then the system shall display developer of the intervention

And where clinically indicated, the system shall display bibliographic citation

Scenario: Display source attributes for linked referential CDS

Given linked referential CDS is being utilized

When I review the reference source information

Then the system shall display developer of the intervention

And where clinically indicated, the system shall display bibliographic citation

Scenario: Access complete source attribute information

Given CDS interventions are configured in the system

When I need to review source attributes

Then all source attributes shall be accessible
And source attributes shall be displayed in user-friendly format
And source attributes shall support evidence-based decision making

6.3 Screenshots

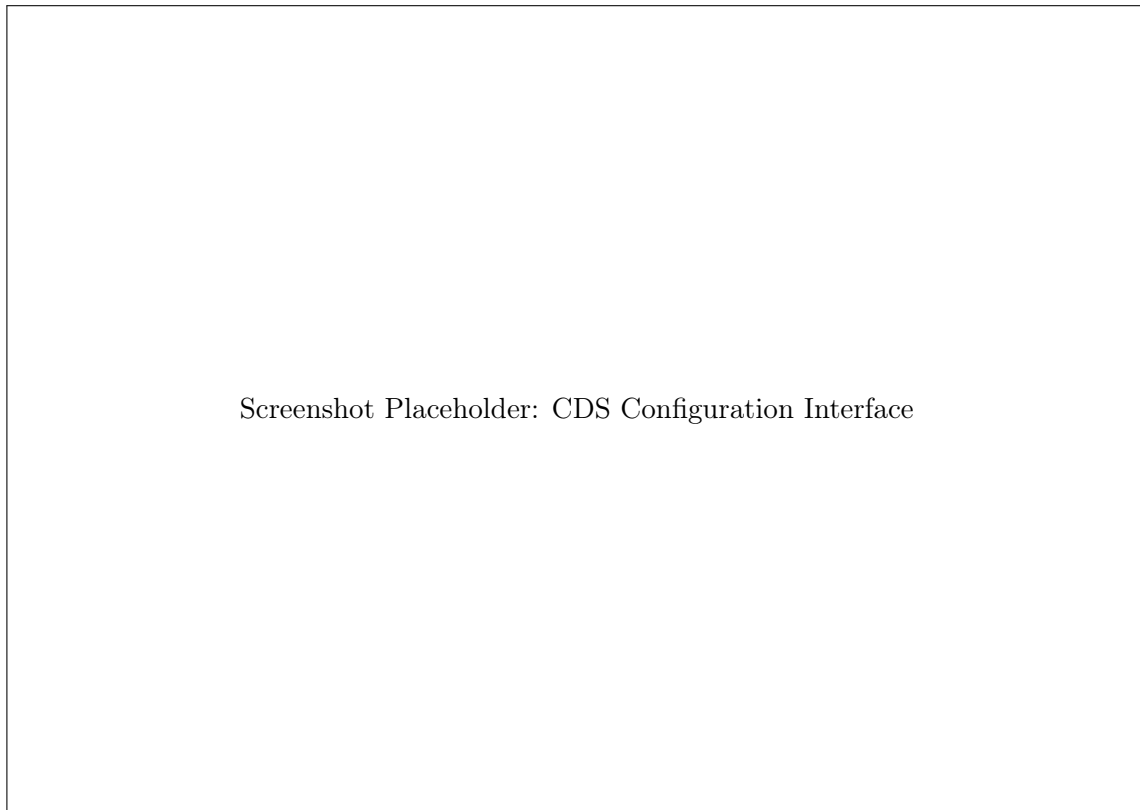


Figure 6.1: Clinical Decision Support - Configuration Interface

Chapter 7

170.315(b)(11) - Decision Support Interventions

7.1 Criterion Information

7.1.1 Maturity Level

IMPLEMENTED

7.1.2 Replacement Notice

This criterion REPLACES 170.315(a)(9) which expired January 1, 2025.

Decision Support Interventions (DSI) is an enhanced version of the legacy Clinical Decision Support criterion, adding requirements for:

- Predictive decision support interventions (AI/ML algorithms)
- Demographic data usage transparency
- Electronic intervention feedback mechanism
- Enhanced risk and fairness disclosures for predictive DSI

7.1.3 Regulatory Text

From *45 CFR 170.315(b)(11)*:

(11) Decision Support Interventions.

(Full regulatory text includes all requirements from a-9 plus enhanced requirements for predictive interventions, demographic transparency, feedback mechanisms, and algorithm governance. See BDD specification for complete requirements.)

7.2 Behavior-Driven Development Specification

Listing 7.1: 170.315(b)(11) - Decision Support Interventions

```
# /Volumes/SonicMagic/Code/honeycomb-public-release/certification/bdd/170.315-b-11-
  decision-support-interventions.feature
# 170.315(b)(11) - Decision Support Interventions

# REGULATORY TEXT FROM 45 CFR 170.315(b)(11)
#
# (11) Use of date of birth as expressed in the standards in 170.213;

@170.315-b-11 @dsi @predictive-ai @algorithm-transparency @care-coordination
Feature: Decision Support Interventions
  As a healthcare provider
  I want transparent and fair decision support interventions
  So that I can make evidence-based decisions while understanding AI/algorithm influences

Background:
  Given I am authenticated as a healthcare provider
  And decision support interventions are configured

# ----- Intervention Interaction -----
```

```
Scenario: Interventions occur during user interaction
  Given I am actively interacting with health IT
  When decision support is triggered
  Then interventions shall be provided during my interaction with technology
  And interventions shall occur at clinically appropriate times

# ----- Configuration -----

Scenario: Configure DSI interventions based on user role
  Given I am authorized to configure decision support
  When configuring interventions
  Then the system shall enable configuration by limited set of identified users
  And configuration shall be based on user role
  And configuration capabilities shall be appropriately restricted

Scenario: Enable interventions when data incorporated from transition of care
  Given medications, allergies, and problems are incorporated from transition summary
  When data is incorporated per 170.315(b)(2)(iii)(D)
  Then interventions shall be enabled based on incorporated data
  And interventions shall trigger appropriately

Scenario: Provide electronic feedback capability for interventions
  Given I am using evidence-based decision support interventions
  When I want to provide feedback on intervention
  Then the system shall enable me to provide electronic feedback data
  And feedback shall be collectible for selected interventions
  And limited set of users shall be able to export feedback in computable format

Scenario: Export intervention feedback data
  Given electronic feedback has been collected
  When authorized users need to analyze feedback
  Then feedback data shall be exportable in computable format
  And export shall include intervention, action taken, user feedback, user, date, and
    location
  And feedback analysis shall support intervention improvement

# ----- Evidence-Based Decision Support Intervention Selection -----

Scenario: Select evidence-based interventions using USCDI data
  Given I am authorized to select decision support interventions
  When activating evidence-based interventions
  Then interventions may use problems per 170.213
  And interventions may use medications per 170.213
  And interventions may use allergies and intolerances per 170.213
  And interventions may use demographics per 170.213
  And interventions may use laboratory data per 170.213
  And interventions may use vital signs per 170.213
  And interventions may use unique device identifiers per 170.213
  And interventions may use procedures per 170.213

# ----- Predictive Decision Support Intervention Selection -----

Scenario: Select predictive interventions using any USCDI data
```

Given I am authorized to select decision support interventions
 When activating predictive decision support interventions
 Then interventions may use any data expressed in 170.213
 And predictive algorithms shall utilize appropriate data elements
 And data inputs shall be transparent

----- Source Attributes for Evidence-Based Interventions -----

Scenario: Display evidence-based intervention source attributes
 Given an evidence-based intervention is active
 When I review intervention source information
 Then the system shall support display of bibliographic citation
 And the system shall support display of intervention developer
 And the system shall support display of funding source for development
 And the system shall support display of release and revision dates

Scenario: Display demographic data usage in evidence-based intervention
 Given an evidence-based intervention uses demographic data
 When I review intervention details
 Then the system shall indicate use of race per 170.213
 And the system shall indicate use of ethnicity per 170.213
 And the system shall indicate use of language per 170.213
 And the system shall indicate use of sexual orientation per 170.213
 And the system shall indicate use of gender identity per 170.213
 And the system shall indicate use of sex per 170.213
 And the system shall indicate use of date of birth per 170.213

Scenario: Display SDOH and health status data usage
 Given an evidence-based intervention uses SDOH data
 When I review intervention details
 Then the system shall indicate use of SDOH data per 170.213
 And the system shall indicate use of health status assessments data per 170.213

----- Source Attributes for Predictive Interventions (EXTENSIVE) -----

Scenario: Display predictive intervention details and output
 Given a predictive decision support intervention is active
 When I review intervention details
 Then the system shall display name and contact information for developer
 And the system shall display funding source of technical implementation
 And the system shall display description of value produced as output
 And the system shall display whether output is prediction, classification, recommendation, evaluation, analysis, or other type

Scenario: Display predictive intervention purpose
 Given a predictive intervention is active
 When I review intervention purpose
 Then the system shall display intended use of intervention
 And the system shall display intended patient population(s)
 And the system shall display intended user(s)
 And the system shall display intended decision-making role (informs, augments, replaces clinical management)

Scenario: Display cautioned out-of-scope use

Given a predictive intervention has usage limitations
When I review intervention limitations
Then the system shall display description of cautioned tasks, situations, or populations
And the system shall display known risks, inappropriate settings, inappropriate uses, or limitations

Scenario: Display intervention development details and input features

Given a predictive intervention is active
When I review development details
Then the system shall display exclusion and inclusion criteria for training data
And the system shall display use of demographic variables as input features
And the system shall display description of demographic representativeness in training data
And the system shall display description of relevance of training data to deployed setting

Scenario: Display fairness process for intervention development

Given a predictive intervention is active
When I review fairness information
Then the system shall display description of approach to ensure intervention output is fair
And the system shall display description of approaches to manage, reduce, or eliminate bias

Scenario: Display external validation process

Given a predictive intervention has been externally validated
When I review validation information
Then the system shall display description of data source, setting, or environment for validation
And the system shall display party that conducted external testing
And the system shall display demographic representativeness of external data
And the system shall display description of external validation process

Scenario: Display quantitative performance measures

Given a predictive intervention is active
When I review performance metrics
Then the system shall display validity in test data from training source
And the system shall display fairness in test data from training source
And the system shall display validity in external data
And the system shall display fairness in external data
And the system shall display references to evaluation of outcomes impact

Scenario: Display ongoing maintenance information

Given a predictive intervention is deployed
When I review maintenance information
Then the system shall display process and frequency for monitoring validity over time
And the system shall display validity in local data
And the system shall display process and frequency for monitoring fairness over time
And the system shall display fairness in local data

Scenario: Display update and validation schedule

Given a predictive intervention is in use
When I review update information

Then the system shall display process and frequency for intervention updates
 And the system shall display frequency for performance correction when risks
 identified

Scenario: Indicate when optional information not available

Given a predictive intervention source attributes
 When certain information is not available for review
 Then the system shall indicate when information not available for external validation
 And the system shall indicate when information not available for external validity
 metrics
 And the system shall indicate when information not available for external fairness
 metrics
 And the system shall indicate when information not available for outcome evaluations
 And the system shall indicate when information not available for local validity
 And the system shall indicate when information not available for local fairness
 And the system shall indicate when information not available for update schedule
 details

----- Access and Modify Source Attributes -----

Scenario: Access complete plain language descriptions for developer-supplied
 interventions

Given interventions are supplied by health IT developer
 When authorized users need to review source attributes
 Then limited set of identified users shall be able to access complete descriptions
 And descriptions shall be in plain language
 And descriptions shall be up-to-date

Scenario: Modify source attributes for interventions

Given intervention source attributes need updating
 When authorized users modify attributes
 Then limited set of identified users shall record, change, and access source
 attributes
 And modifications shall be tracked with audit trail
 And all users shall see updated information

Scenario: Record additional attributes for predictive interventions

Given predictive interventions may have additional relevant attributes
 When documenting intervention characteristics
 Then limited set of users shall record, change, and access additional attributes
 And additional attributes supplement required attributes
 And comprehensive documentation shall be maintained

----- Intervention Risk Management for Predictive DSI -----

Scenario: Conduct risk analysis for predictive intervention

Given a predictive decision support intervention is implemented
 When assessing intervention risks
 Then intervention shall be subject to analysis of potential risks and adverse impacts
 And analysis shall assess validity, reliability, robustness, fairness
 And analysis shall assess intelligibility, safety, security, and privacy

Scenario: Implement risk mitigation for predictive intervention

Given risks have been identified for predictive intervention

When implementing risk controls

Then intervention shall be subject to practices to mitigate identified risks

And mitigation shall address all significant risk categories

And mitigation effectiveness shall be monitored

Scenario: Apply governance to predictive intervention

Given a predictive intervention is in production use

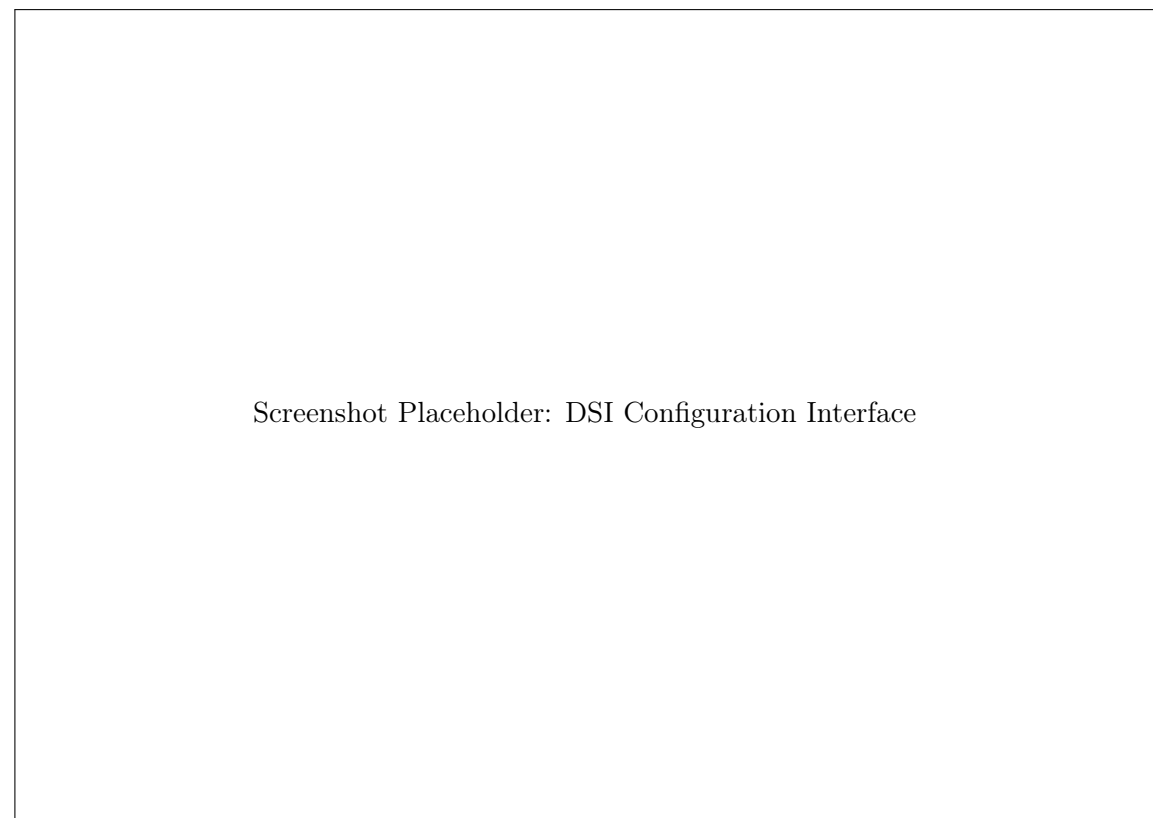
When managing intervention lifecycle

Then intervention shall be subject to policies and implemented controls for governance

And governance shall include how data are acquired, managed, and used

And governance shall ensure ongoing oversight and accountability

7.3 Screenshots



Screenshot Placeholder: DSI Configuration Interface

Figure 7.1: Decision Support Interventions - Configuration Interface

Chapter 8

170.315(a)(14) - Implantable Device List

8.1 Criterion Information

8.1.1 Maturity Level

IMPLEMENTED

8.1.2 Regulatory Text

From 45 CFR 170.315(a)(14):

(14) Implantable device list.

(i) Record Unique Device Identifiers associated with a patient's Implantable Devices.

(ii) Parse the following identifiers from a Unique Device Identifier.

(Full regulatory text abbreviated. See BDD specification for complete requirements.)

8.2 Behavior-Driven Development Specification

Listing 8.1: 170.315(a)(14) - Implantable Device List

```
# /Volumes/SonicMagic/Code/honeycomb-public-release/certification/bdd/170.315-a-14-
  implantable-device-list.feature
# 170.315(a)(14) - Implantable Device List

# REGULATORY TEXT FROM 45 CFR 170.315(a)(14)
#
# (14) Implantable device list.
#
# (i) Record Unique Device Identifiers associated with a patient's Implantable Devices.
#
# (ii) Parse the following identifiers from a Unique Device Identifier:
#
# (A) Device Identifier; and
#
# (B) The following identifiers that compose the Production Identifier:
#
# (1) The lot or batch within which a device was manufactured;
#
# (2) The serial number of a specific device;
#
# (3) The expiration date of a specific device;
#
# (4) The date a specific device was manufactured; and
#
# (5) For an HCT/P regulated as a device, the distinct identification code required by 21
    CFR 1271.290(c).
#
# (iii) Obtain and associate with each Unique Device Identifier:
#
# (A) A description of the implantable device referenced by at least one of the following
    :
#
```

```

# (1) The "GMDN PT Name" attribute associated with the Device Identifier in the Global
    Unique Device Identification Database.
#
# (2) The "SNOMED CT Description" mapped to the attribute referenced in in paragraph (a)
    (14)(iii)(A)(1) of this section.
#
# (B) The following Global Unique Device Identification Database attributes:
#
# (1) "Brand Name";
#
# (2) "Version or Model";
#
# (3) "Company Name";
#
# (4) "What MRI safety information does the labeling contain?"; and
#
# (5) "Device required to be labeled as containing natural rubber latex or dry natural
    rubber (21 CFR 801.437). "
#
# (iv) Display to a user an implantable device list consisting of:
#
# (A) The active Unique Device Identifiers recorded for the patient;
#
# (B) For each active Unique Device Identifier recorded for a patient, the description of
    the implantable device specified by paragraph (a)(14)(iii)(A) of this section; and
#
# (C) A method to access all Unique Device Identifiers recorded for a patient.
#
# (v) For each Unique Device Identifier recorded for a patient, enable a user to access:
#
# (A) The Unique Device Identifier;
#
# (B) The description of the implantable device specified by paragraph (a)(14)(iii)(A) of
    this section;
#
# (C) The identifiers associated with the Unique Device Identifier, as specified by
    paragraph (a)(14)(ii) of this section; and
#
# (D) The attributes associated with the Unique Device Identifier, as specified by
    paragraph (a)(14)(iii)(B) of this section.
#
# (vi) Enable a user to change the status of a Unique Device Identifier recorded for a
    patient.

@170.315-a-14 @udi @implantable-devices @clinical
Feature: Implantable Device List
  As a healthcare provider
  I want to record and track implantable devices using Unique Device Identifiers
  So that I can ensure patient safety, support recall management, and provide accurate
    device information

Background:
  Given I am authenticated as a healthcare provider
  And I have a patient record open

```

----- Record UDI -----

Scenario: Record Unique Device Identifier for implantable device

Given a patient has received an implantable device

When I record the device information

Then the system shall enable recording of Unique Device Identifiers

And the UDI shall be associated with the patient record

And the UDI shall be stored in structured format

----- Parse UDI Components -----

Scenario: Parse Device Identifier from UDI

Given a Unique Device Identifier has been recorded

When the system processes the UDI

Then the system shall parse the Device Identifier (DI)

And the DI shall be stored as separate data element

Scenario: Parse Production Identifier components from UDI

Given a Unique Device Identifier has been recorded

When the system processes the UDI

Then the system shall parse lot or batch number

And the system shall parse serial number

And the system shall parse expiration date

And the system shall parse manufacture date

And all production identifiers shall be stored as separate data elements

Scenario: Parse HCT/P distinct identification code from UDI

Given a Unique Device Identifier for HCT/P regulated device has been recorded

When the system processes the UDI

Then the system shall parse distinct identification code per 21 CFR 1271.290(c)

And the code shall be stored appropriately

----- Obtain Device Information from GUDID -----

Scenario: Obtain device description from GUDID

Given a Unique Device Identifier has been entered

When the system queries the Global Unique Device Identification Database

Then the system shall obtain device description

And description may be "GMDN_PT_Name" from GUDID

Or description may be "SNOMED_CT_Description" mapped to GMDN PT Name

Scenario: Obtain device brand name from GUDID

Given a Unique Device Identifier has been entered

When the system queries GUDID

Then the system shall obtain "Brand_Name" attribute

And the brand name shall be associated with the UDI

Scenario: Obtain device version or model from GUDID

Given a Unique Device Identifier has been entered

When the system queries GUDID

Then the system shall obtain "Version_or_Model" attribute

And the version/model shall be associated with the UDI

Scenario: Obtain device company name from GUDID
Given a Unique Device Identifier has been entered
When the system queries GUDID
Then the system shall obtain "Company_Name" attribute
And the manufacturer information shall be associated with the UDI

Scenario: Obtain MRI safety information from GUDID
Given a Unique Device Identifier has been entered
When the system queries GUDID
Then the system shall obtain MRI safety information
And the information shall indicate MRI labeling content
And MRI safety information shall be prominently available

Scenario: Obtain latex allergy information from GUDID
Given a Unique Device Identifier has been entered
When the system queries GUDID
Then the system shall obtain latex labeling information per 21 CFR 801.437
And latex information shall be prominently available for patient safety

----- *Display Implantable Device List* -----

Scenario: Display patient's implantable device list
Given a patient has recorded Unique Device Identifiers
When I view the implantable device list
Then the system shall display active Unique Device Identifiers
And the system shall display device descriptions
And the display shall be organized and easily reviewable

Scenario: Display device description for each active UDI
Given a patient has multiple active implantable devices
When I view the implantable device list
Then for each active UDI the system shall display device description
And description shall be from GUDID per requirements
And description shall be clinically meaningful

Scenario: Provide method to access all UDIs for patient
Given a patient has both active and inactive device records
When I need to view complete device history
Then the system shall provide method to access all UDIs
And I shall be able to distinguish active from inactive devices
And complete device history shall be accessible

----- *Access Individual UDI Details* -----

Scenario: Access complete UDI information
Given a patient has a recorded Unique Device Identifier
When I select a specific UDI to review
Then the system shall enable access to the complete UDI
And the system shall display the device description
And the system shall display parsed production identifiers
And the system shall display GUDID attributes

Scenario: Access parsed identifiers for UDI
Given a patient has a recorded Unique Device Identifier

```
When I access the UDI details
Then the system shall display lot/batch number
And the system shall display serial number
And the system shall display expiration date
And the system shall display manufacture date
And the system shall display HCT/P code if applicable
```

Scenario: Access GUDID attributes for UDI

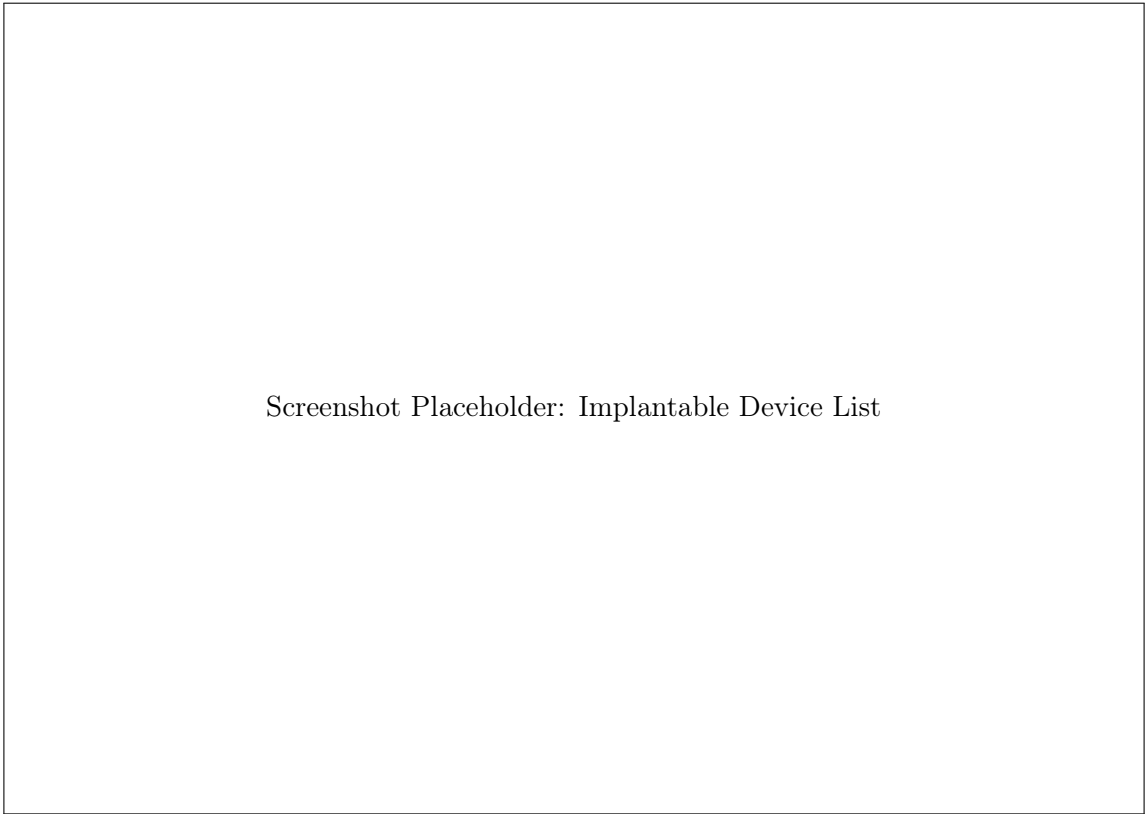
```
Given a patient has a recorded Unique Device Identifier
When I access the UDI details
Then the system shall display brand name
And the system shall display version or model
And the system shall display company name
And the system shall display MRI safety information
And the system shall display latex information
```

----- *Change UDI Status* -----

Scenario: Change status of implantable device

```
Given a patient has a recorded Unique Device Identifier
When the device is explanted or status changes
Then the system shall enable changing the status of the UDI
And status change shall be recorded with date and reason
And device shall move from active to inactive list
```

8.3 Screenshots



Screenshot Placeholder: Implantable Device List

Figure 8.1: Implantable Device List - Patient View

Chapter 9

170.315(e)(1) - View, Download, Transmit

9.1 Criterion Information

9.1.1 Maturity Level

IMPLEMENTED

9.1.2 Regulatory Text

From 45 CFR 170.315(e)(1):

(1) View, download, and transmit to 3rd party.

The technology must enable patients to view, download, and transmit their health information.

(Full regulatory text abbreviated. See BDD specification for complete requirements.)

9.2 Behavior-Driven Development Specification

Listing 9.1: 170.315(e)(1) - View, Download, Transmit

```
# /Volumes/SonicMagic/Code/honeycomb-public-release/certification/bdd/170.315-e-1-view-
  download-transmit.feature
# 170.315(e)(1) - View, Download, and Transmit to 3rd Party

# REGULATORY TEXT FROM 45 CFR 170.315(e)(1)
#
# (1) The action(s) (i.e., view, download, transmission) that occurred;

@170.315-e-1 @vdt @patient-access @patient-engagement
Feature: View, Download, and Transmit to 3rd Party
  As a patient
  I want to view, download, and transmit my health information
  So that I can access my data and share it with others

  Background:
    Given I am authenticated as a patient
    And my health information is available in the system

  Scenario: View health information online
    Given I am accessing my patient portal
    When I view my health information
    Then the system shall display my data in human readable format
    And all available USCDI data shall be viewable
    And the information shall be current and accurate

  Scenario: Download health information
    Given I am viewing my health information
    When I choose to download my data
    Then the system shall enable download
    And download shall be in machine-readable format
    And download shall include all requested data

  Scenario: Transmit to third party via encrypted email
```

Given I want to share my health information
When I transmit to a third party email address
Then the system shall transmit encrypted
And transmission shall be secure
And third party shall be able to receive the data

Scenario: Transmit to third party unencrypted with patient consent

Given I want to share my health information
And I accept the risks of unencrypted transmission
When I choose unencrypted transmission
Then the system shall warn me of risks
And system shall transmit unencrypted only with my consent
And my choice shall be documented

Scenario: Access activity history log

Given I have viewed, downloaded, or transmitted my data
When I check my activity history
Then the system shall display activity log
And log shall show what data was accessed
And log shall show when data was accessed
And log shall show to whom data was transmitted

9.3 Screenshots

Screenshot Placeholder: Patient Portal VDT Interface

Figure 9.1: View, Download, Transmit - Patient Portal Interface

Chapter 10

170.315(g)(7) - Application Access - Patient Selection

10.1 Criterion Information

10.1.1 Maturity Level

IMPLEMENTED

10.1.2 Regulatory Text

From 45 CFR 170.315(g)(7):

(7) Application accesspatient selection.

The technology must be able to receive a request with sufficient information to uniquely identify a patient and return an ID or other token.

(Full regulatory text abbreviated. See BDD specification for complete requirements.)

10.2 Behavior-Driven Development Specification

Listing 10.1: 170.315(g)(7) - Application Access

```
# /Volumes/SonicMagic/Code/honeycomb-public-release/certification/bdd/170.315-g-7-10-apis
# .feature
# 170.315(g)(7)-(g)(10) - Application Programming Interfaces

# REGULATORY TEXT FROM 45 CFR 170.315(g)(7)
#
# (7) Application accesspatient selection. The following technical outcome and conditions
# must be met through the demonstration of an application programming interface (API).
# (i) Functional requirement. The technology must be able to receive a request with
# sufficient information to uniquely identify a patient and return an ID or other token
# that can be used by an application to subsequently execute requests for that patient
# 's data.
# (ii) Documentation.
# (A) The API must include accompanying documentation that contains, at a minimum:

@170.315-g-7-10 @api @fhir @design-performance
Feature: Application Programming Interfaces
  As a health IT developer
  I want to provide standardized APIs
  So that applications can access health information

Background:
  Given the Health IT Module provides APIs
  And API access is required

Scenario: Provide patient selection API (g)(7)
  Given applications need to select patients
  When application requests patient selection
  Then API shall enable patient selection
  And API shall conform to standards
  And API shall be properly documented

Scenario: Provide data category request API (g)(8)
```

Given applications need specific data categories
When application requests data by category
Then API shall provide data category request capability
And API shall return requested data
And API shall conform to USCDI

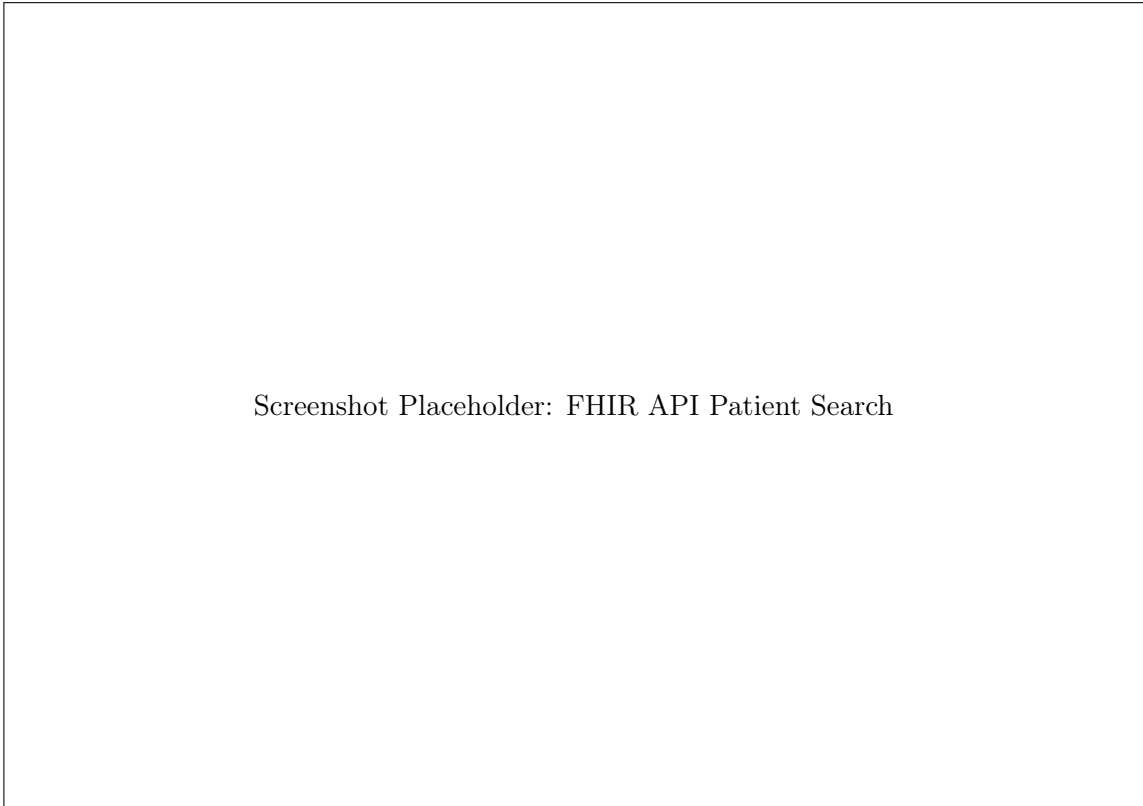
Scenario: Provide all data request API (g)(9)

Given applications need complete patient data
When application requests all patient data
Then API shall provide all available data
And data shall include all USCDI elements
And API shall conform to standards

Scenario: Provide standardized API for patient and population services (g)(10)

Given applications need patient and population data
When application uses FHIR API
Then API shall conform to 170.215(a) standard
And API shall support SMART App Launch
And API shall support bulk data access
And API shall provide comprehensive functionality

10.3 Screenshots



Screenshot Placeholder: FHIR API Patient Search

Figure 10.1: Application Access - Patient Selection API

Chapter 11

170.315(g)(9) - Application Access - All Data

11.1 Criterion Information

11.1.1 Maturity Level

IMPLEMENTED

11.1.2 Regulatory Text

From 45 CFR 170.315(g)(9):

(9) Application accessall data request.

Provide an application programming interface (API) that can return all available data for a specific patient in a single request.

(Full regulatory text abbreviated. See BDD specification for complete requirements.)

11.2 Behavior-Driven Development Specification

Listing 11.1: 170.315(g)(9) - Application Access

```
# /Volumes/SonicMagic/Code/honeycomb-public-release/certification/bdd/170.315-g-7-10-apis
.feature
# 170.315(g)(7)-(g)(10) - Application Programming Interfaces

# REGULATORY TEXT FROM 45 CFR 170.315(g)(7)
#
# (7) Application accesspatient selection. The following technical outcome and conditions
# must be met through the demonstration of an application programming interface (API).
# (i) Functional requirement. The technology must be able to receive a request with
# sufficient information to uniquely identify a patient and return an ID or other token
# that can be used by an application to subsequently execute requests for that patient
# 's data.
# (ii) Documentation.
# (A) The API must include accompanying documentation that contains, at a minimum:

@170.315-g-7-10 @api @fhir @design-performance
Feature: Application Programming Interfaces
  As a health IT developer
  I want to provide standardized APIs
  So that applications can access health information

Background:
  Given the Health IT Module provides APIs
  And API access is required

Scenario: Provide patient selection API (g)(7)
  Given applications need to select patients
  When application requests patient selection
  Then API shall enable patient selection
  And API shall conform to standards
  And API shall be properly documented

Scenario: Provide data category request API (g)(8)
```

Given applications need specific data categories
When application requests data by category
Then API shall provide data category request capability
And API shall return requested data
And API shall conform to USCDI

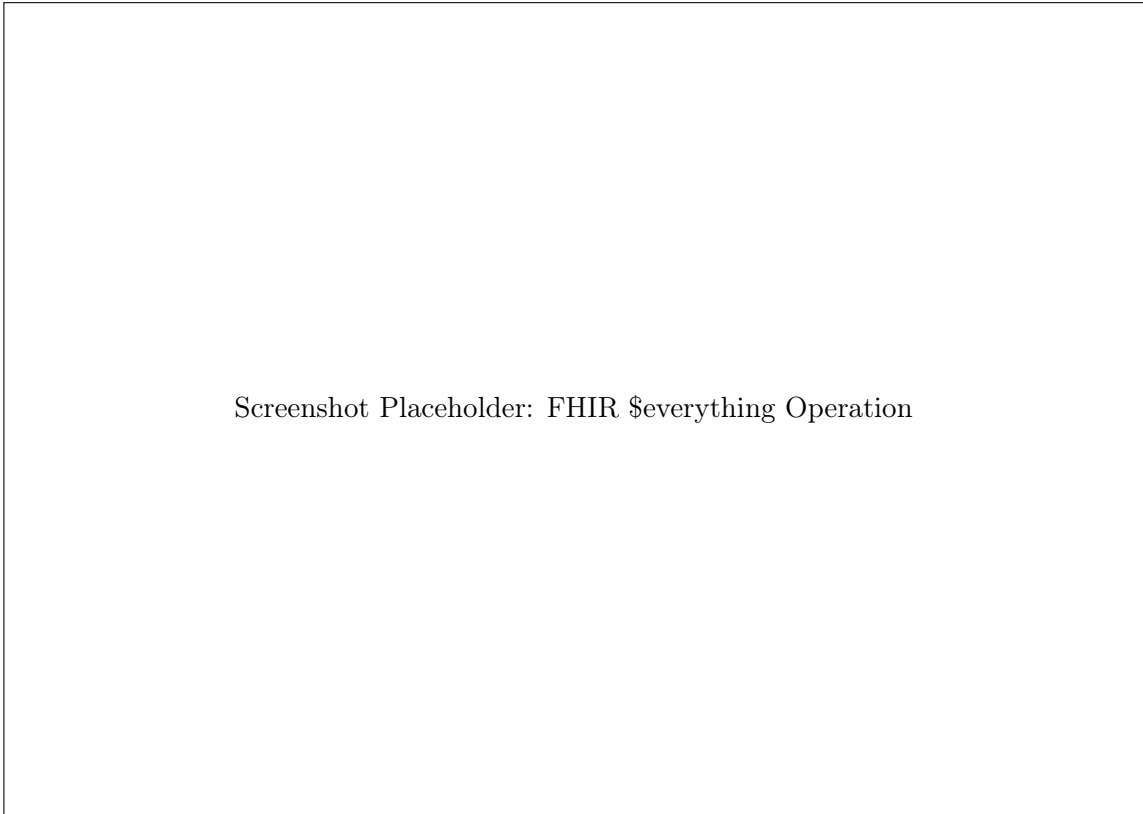
Scenario: Provide all data request API (g)(9)

Given applications need complete patient data
When application requests all patient data
Then API shall provide all available data
And data shall include all USCDI elements
And API shall conform to standards

Scenario: Provide standardized API for patient and population services (g)(10)

Given applications need patient and population data
When application uses FHIR API
Then API shall conform to 170.215(a) standard
And API shall support SMART App Launch
And API shall support bulk data access
And API shall provide comprehensive functionality

11.3 Screenshots



Screenshot Placeholder: FHIR \$everything Operation

Figure 11.1: Application Access - All Data Request API

Chapter 12

170.315(g)(10) - Standardized API

12.1 Criterion Information

12.1.1 Maturity Level

IMPLEMENTED

12.1.2 Regulatory Text

From 45 CFR 170.315(g)(10):

(10) Standardized API for patient and population services.

The technology must implement a standardized API that conforms to FHIR R4, US Core, SMART App Launch, OpenID Connect, and SMART Backend Services.

(Full regulatory text abbreviated. See BDD specification for complete requirements.)

12.2 Behavior-Driven Development Specification

Listing 12.1: 170.315(g)(10) - Standardized API

```
# /Volumes/SonicMagic/Code/honeycomb-public-release/certification/bdd/170.315-g-7-10-apis
# .feature
# 170.315(g)(7)-(g)(10) - Application Programming Interfaces

# REGULATORY TEXT FROM 45 CFR 170.315(g)(7)
#
# (7) Application accesspatient selection. The following technical outcome and conditions
# must be met through the demonstration of an application programming interface (API).
# (i) Functional requirement. The technology must be able to receive a request with
# sufficient information to uniquely identify a patient and return an ID or other token
# that can be used by an application to subsequently execute requests for that patient
# 's data.
# (ii) Documentation.
# (A) The API must include accompanying documentation that contains, at a minimum:

@170.315-g-7-10 @api @fhir @design-performance
Feature: Application Programming Interfaces
  As a health IT developer
  I want to provide standardized APIs
  So that applications can access health information

Background:
  Given the Health IT Module provides APIs
  And API access is required

Scenario: Provide patient selection API (g)(7)
  Given applications need to select patients
  When application requests patient selection
  Then API shall enable patient selection
  And API shall conform to standards
  And API shall be properly documented

Scenario: Provide data category request API (g)(8)
```

Given applications need specific data categories
When application requests data by category
Then API shall provide data category request capability
And API shall return requested data
And API shall conform to USCDI

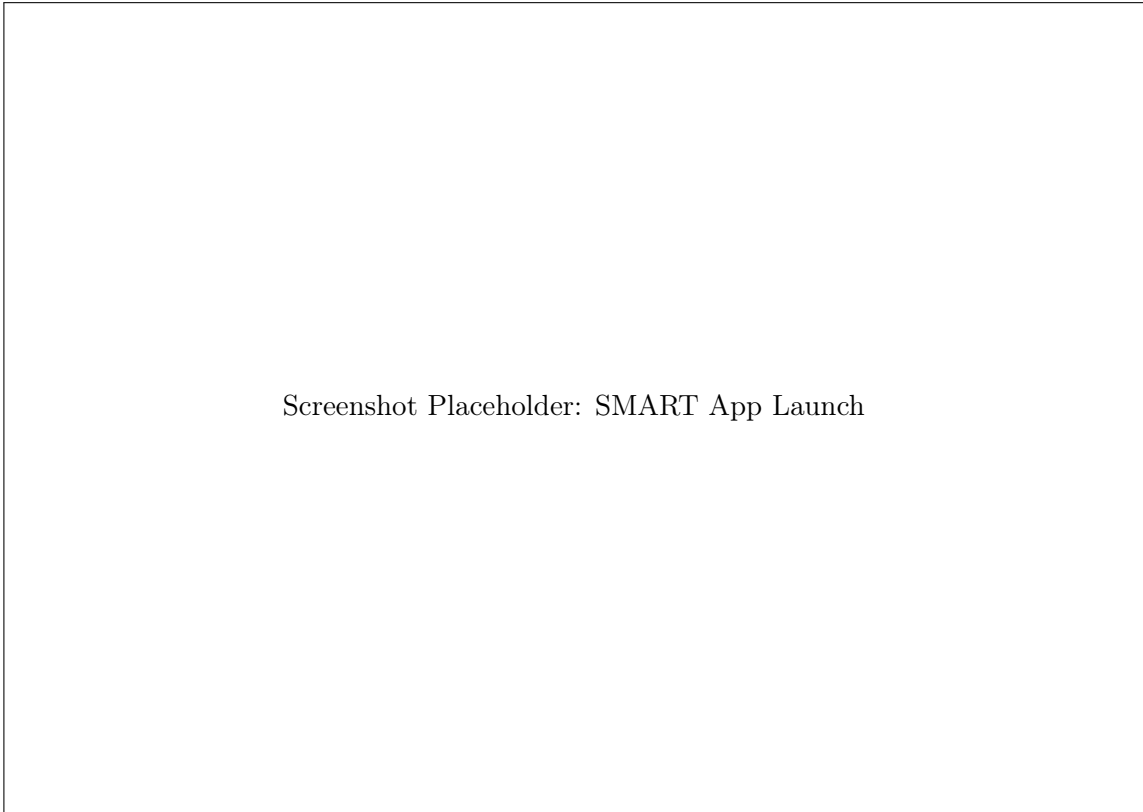
Scenario: Provide all data request API (g)(9)

Given applications need complete patient data
When application requests all patient data
Then API shall provide all available data
And data shall include all USCDI elements
And API shall conform to standards

Scenario: Provide standardized API for patient and population services (g)(10)

Given applications need patient and population data
When application uses FHIR API
Then API shall conform to 170.215(a) standard
And API shall support SMART App Launch
And API shall support bulk data access
And API shall provide comprehensive functionality

12.3 Screenshots



Screenshot Placeholder: SMART App Launch

Figure 12.1: Standardized API - SMART App Launch Authorization

Appendix A

Acronyms and Abbreviations

API

Application Programming Interface

BDD

Behavior-Driven Development

C-CDA

Consolidated Clinical Document Architecture

CDS

Clinical Decision Support

CFR

Code of Federal Regulations

CPOE

Computerized Provider Order Entry

DI Device Identifier

DSI Decision Support Interventions

EHR

Electronic Health Record

FHIR

Fast Healthcare Interoperability Resources

GMDN

Global Medical Device Nomenclature

GUDID

Global Unique Device Identification Database

HCT/P

Human Cells, Tissues, and Cellular and Tissue-Based Products

HL7

Health Level Seven

MRI

Magnetic Resonance Imaging

OAuth

Open Authorization

ONC

Office of the National Coordinator for Health Information Technology

PI Production Identifier**SMART**

Substitutable Medical Applications, Reusable Technologies

SNOMED CT

Systematized Nomenclature of Medicine - Clinical Terms

SPCU

Sex Parameter for Clinical Use

UDI

Unique Device Identifier

USCDI

United States Core Data for Interoperability

VDT

View, Download, Transmit

Appendix B

References

1. Office of the National Coordinator for Health Information Technology. *2015 Edition Health IT Certification Criteria, 2015 Edition Base EHR Definition, and ONC Health IT Certification Program Modifications*. 45 CFR Part 170. Federal Register, 2015.
2. Office of the National Coordinator for Health Information Technology. *21st Century Cures Act: Interoperability, Information Blocking, and the ONC Health IT Certification Program*. 45 CFR Part 170. Federal Register, 2020.
3. HL7 International. *FHIR R4 Specification*. <http://hl7.org/fhir/R4/>
4. HL7 International. *US Core Implementation Guide*. <http://hl7.org/fhir/us/core/>
5. SMART Health IT. *SMART App Launch Framework*. <http://hl7.org/fhir/smart-app-launch/>
6. Office of the National Coordinator for Health Information Technology. *United States Core Data for Interoperability (USCDI) Version 5*. 2024.
7. FDA. *Unique Device Identification System (UDI System)*. 21 CFR Part 801.
8. FDA. *Global Unique Device Identification Database (GUDID)*. <https://accessgudid.nlm.nih.gov/>

Appendix C

Certification Testing Notes

C.1 Inferno Testing

For comprehensive certification testing of API criteria (g-7, g-9, g-10), the official ONC-approved Inferno testing tool must be used:

Inferno Website: <https://inferno.healthit.gov>

Inferno provides authoritative test suites that validate:

- FHIR R4 conformance
- US Core Implementation Guide compliance
- SMART App Launch OAuth 2.0 flows
- OpenID Connect Core
- SMART Backend Services (Bulk FHIR)
- USCDI data completeness
- Token management and revocation
- All mandatory and must-support elements

The Nightwatch tests in this system provide basic endpoint verification for development and CI/CD purposes. They are **not** a replacement for Inferno certification testing.

C.2 Test Execution

To run the complete Base EHR test suite:

```
cd /path/to/honeycomb-public-release
chmod +x certification/tdd/base_ehr/run-base-ehr-tests.sh
./certification/tdd/base_ehr/run-base-ehr-tests.sh
```

This script runs all 12 tests (11 base + 1 alternate for a-9/b-11 uncertainty).